EART26201 | Principles of Geochemistry

Week 5 Summative Assessment FEEDBACK

Instructions

This assessment contains two questions that should each take about 60 mins to complete.

Please provide your answers in a single Word document or pdf, copying tables and figures from Excel as appropriate.

Question 1: Pyroxene crystallisation **[50 marks]**

Pyroxene minerals commonly crystallise from basaltic magmas. Assuming that basaltic magmas have a composition of X_{Di} = 0.7, use the phase diagram below to explain how *both* thermodynamics (equilibrium processes) and kinetics (disequilibrium processes) affect the composition, abundance and texture of pyroxenes in rocks formed at different cooling rates. Use the phase diagram below to provide as much quantitative detail as possible. [50 marks]

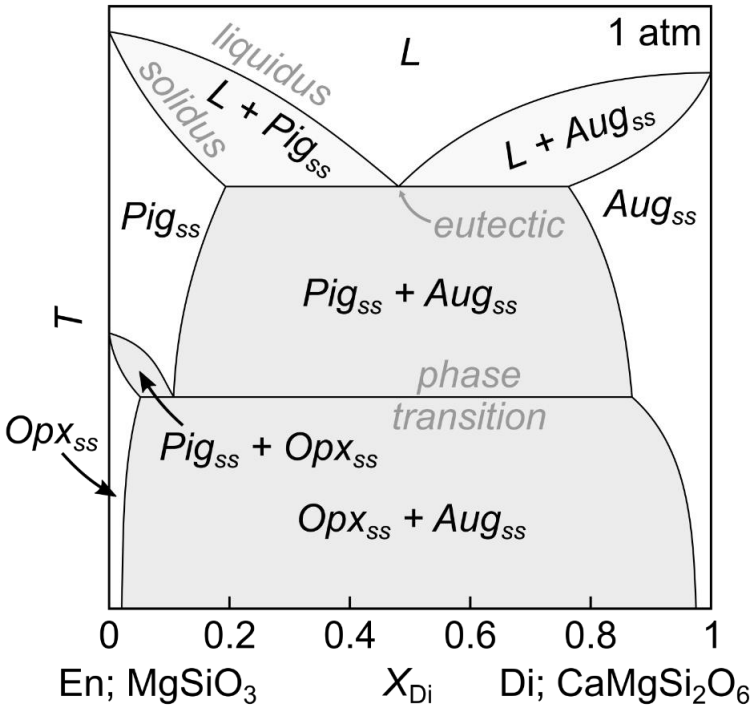


Figure 1: Pyroxene phase diagram

Thermodynamic describes the equilibrium state of a system under a given set of intrinsic conditions (pressure, temperature and composition). The equilibrium state is defined by the phase or phase assemblage that has the lowest Gibbs free energy under the conditions of interest. Phase diagrams provide a way of visualising how the equilibrium state (i.e. phase assemblage) changes as conditions are changed, though are restricted to looking at changes in only two conditions at a time because of graphical limitations; phase diagrams are typically isochemical, isothermal or isobaric projections. However, reaching equilibrium takes time as reactants will typically need to move within the system, whether by advection or diffusion, in order for reactions to occur. As a result, phase diagrams may not adequately describe the behaviour of chemical systems that are subject to kinetic limitations.

When basalt cools slowly in large magma reservoirs, chemical equilibrium is maintained throughout crystallisation and kinetic effects are absent. The first crystal to form from liquid basalt with a composition XDi=0.7 will be augite with a composition of XDi~0.92. As the system cools, both the augite and the liquid will become progressively poorer in Di poor until the liquid reaches the eutectic, at which point the composition of the liquid will be XDi~0.45 and the composition of augite will be XDi~0.77. Eutectic crystallisation will then produce augite (XDi~0.77) and pigeonite (XDi~0.17) in roughly equal proportions until the system is solid and contains about ~90% augite and ~10% pigeonite. Because equilibrium is maintained throughout cooling, Di-poor pigeonite will progressively exsolve from Di-rich augite and vice versa, producing exsolution lamellae. This exsolution will continue until the pigeonite-orthopyroxene phase transition is reached. Before the phase transition, the system will contain about ~78% augite and ~22% pigeonite with compositions of XDi~0.88 and XDi~0.10, respectively; after it will contain ~80% augite and ~20% orthopyroxene with compositions of XDi~0.88 and XDi~0.05, respectively. As the system cools further exsolution will continue, with orthopyroxene eventually reaching a very Di-poor composition of XDi~0.02 and augite reaching a Di-rich composition of XDi~0.98; the system will contain ~~ 29% orthopyroxene. A slowly cooled basalt will thus eventually contain augite with large exsolution lamellae of orthopyroxene as well as orthopyroxenes with large exsolution lamellae of augite (so-called inverted pigeonites).

When basalt cools quickly in lava flows, chemical equilibrium is not maintained because there are kinetic limitations on moving reactants around. The initial crystallisation of a quickly cooling basalt is much the same as that described above for a slowly cooling basalt up to the point of solidification at the eutectic. This is largely because diffusion rates are considerably faster in liquids than solids. However, once the system is solid, diffusion rates may be too slow to move Ca around within crystals limiting the amount of exsolution that can occur or preventing it entirely. Phase transitions will also not occur, preventing the reaction of pigeonite to orthopyroxene. Very quickly cooled basalts will contain augite and pigeonite with eutectic compositions of XDi~0.77 and XDi~0.17 in proportions of ~90% augite and ~10% pigeonite, respectively. Overall, quickly cooled basalts will contain homogenous crystals of augite and pigeonite, without exsolution lamellae.

Question 2: The Sgurr of Eigg **[50 marks]**

Eigg is a small island in the Inner Hebrides of Scotland where a large number of glassy rhyolite dykes outcrop. One of these is called the Sgurr of Eigg.

a) Knowing that $\lambda_{(^{87}\mathrm{Rb})} = 1.42 \times 10^{-11}$ years and that the geochronology equation has the form $\frac{D}{S} = \left(\frac{D}{S}\right)_0 + \frac{N}{S}\left(e^{\lambda t} - 1\right)$, use the data in Table 1 to plot an isochron diagram and calculate the age of the Sgurr of Eigg. Show your working [15 marks]

Sample	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$
ES1 whole-rock	14.65	0.72370
ES1 feldspar	1.79	0.71414
ES1 glass	34.97	0.73928
ES3 whole-rock	9.38	0.71973
ES3 feldspar	1.29	0.71383
ES4 whole-rock	10.63	0.72070

Table 1: Rb-Sr systematics of the Sgurr of Eigg

See Excel for isochron diagram. The age of the Sgurr of Eigg is 56 Ma.

b) Describe how the different partitioning behaviour of parent and daughter isotopes in the Rb-Sr isotope system leads to the creation of isochrons. Use information provided in Table 1 and/or your answer to a) to substantiate your answer. [15 marks]

The parent isotope, 87Rb, is more incompatible than the daughter and stable isotopes 87Sr and 86Sr. Glass compositions are therefore relatively enriched in 87Rb with respect to feldspar compositions because glass represent frozen liquid that was enriched in incompatible elements during crystallisation. Whole-rock compositions lie between glass and feldspar compositions because they are a mixture of the two. As time progresses, high glass 87Rb contents translate into high 87Sr contents. The greater the amount of radioactive parent isotope, the greater the amount of daughter isotope produced after any given time. Correspondingly, low feldspar 87Rb contents will result in low 87Sr contents. Glass and feldspar thus define two ends of an isochron, the former being rich in both parent 87Rb and daughter 87 Sr isotopes with respect to the stable 86Sr isotope, and the latter being poor in both.

c) Determine the initial Sr isotope ratio, $(^{87}Sr/^{86}Sr)_0$, of the Sgurr of Eigg. [5 marks]

0.7127

d) Use your answer to c), the initial Nd isotope ratio of the Sgurr of Eigg (0.51146), and the data in Table 2 to plot $(^{87}Sr/^{86}Sr)_0$ versus $(^{143}Nd/^{144}Nd)_0$ for the Sgurr of Eigg and its possible magma sources. Using this plot, discussed magma source or sources from which the Sgurr of Eigg mostly likely originated? [15 marks]

Possible magma sources	$(^{87}Sr/^{86}Sr)_0$	$(^{143}Nd/^{144}Nd)_0$
Mantle	0.7028	0.51300
Torridonian sandstone	0.7406	0.51178
Lewisian amphibolite	0.7174	0.51082
Lewisian granulite	0.7030	0.51082

Table 2: Isotope systematics of potential magma sources

See Excel for scatter plot diagram. The combined Sd and Nd isotope composition of the Sgurr of Eigg lies between the high-Nd mantle end-member and the low-Nd crustal end members. The composition cannot be created from mixing the crustal end-members in any combination, so at least some mantle input is required. Overall, the composition lies between values for the mantle and Lewisian amphibolite. Contributions from either Lewisian granulite of Torridonian sandstone would require contribution from the other in order to balance our their low and high 87Sr/87Sr characters, respectively, making this scenario less likely than the simple mixing described above.